

Project: BizKredit

SME Business Loan & Working Capital Platform

Domain: Banking & Financial Services

1. Introduction

BizKredit is a web-based SME business loan and working capital platform designed for commercial banks, SME-focused NBFCs, and fintech lenders to manage end-to-end business loan origination, financial statement analysis, credit underwriting, collateral evaluation, loan disbursement, trade finance-linked working capital facilities, and SME portfolio performance monitoring. The system enables SME applicants to submit business loan applications, credit analysts to assess financial health and approve facilities, relationship managers to manage SME client portfolios, and administrators to configure loan products, underwriting scorecards, and covenant rules.

It uses a REST API-based backend architecture with Angular/React frontend and is compatible with Java (Spring Boot) and .NET (ASP.NET Core) platforms.

Actors / Users

- **SME Applicant:** Submits loan applications, uploads business documents, and tracks application status
- **Credit Analyst:** Reviews financial statements, computes credit metrics, and prepares credit proposals
- **Underwriting Manager:** Approves or declines credit proposals and sets facility terms and conditions
- **Relationship Manager:** Manages SME client portfolio, facility renewals, and covenant monitoring
- **Collateral Evaluator:** Assesses and registers collateral assets with valuation details
- **SME Credit Admin:** Configures loan products, underwriting scorecards, covenant templates, and user roles

2. Module Overview

- **2.1 Identity & Access Management:** Authentication, RBAC, and audit trails.
- **2.2 SME Applicant & Business Profile Management:** Manages SME business profiles, KYC, and ownership structure.

- **2.3 Loan Application & Document Management:** Handles loan applications, required document checklists, and submission tracking.
- **2.4 Financial Analysis & Credit Underwriting:** Processes financial statements, computes credit ratios, and prepares credit proposals.
- **2.5 Collateral Management:** Registers, values, and tracks collateral assets linked to SME facilities.
- **2.6 Facility Disbursement & Working Capital Management:** Manages facility sanctioning, drawdowns, repayment, and working capital utilisation tracking.
- **2.7 Covenant & Portfolio Monitoring:** Tracks financial covenants, early warning signals, and SME portfolio health.
- **2.8 Notifications & Alerts:** Application status updates, covenant breach warnings, facility renewal reminders, and repayment alerts.

3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST API-based architecture
- **Database:** Relational Database (MySQL / PostgreSQL / SQL Server)

4. Module-Wise Design

4.1 Identity & Access Management Module

Features

- SME applicant and bank staff registration with role-based access control
- Maker-checker enforcement on credit proposals and full audit logging

Entities

- **User**
 - UserID
 - Name
 - Role
(SMEApplicant/CreditAnalyst/UnderwritingManager/RelationshipManager/CollateralEvaluator/Admin)
 - Email
 - Phone

- BranchID
- Status (Active/Locked/Inactive)
- **AuditLog**
 - AuditID
 - UserID
 - Action
 - EntityType
 - RecordID
 - Timestamp

4.2 SME Applicant & Business Profile Management Module

Features

- Register SME business entities with ownership, industry, and operational details
- Manage directors, promoters, and group company linkages for consolidated assessment

Entities

- **SMEBusiness**
 - BusinessID
 - BusinessName
 - RegistrationNumber
 - EntityType (PrivateLimited/Partnership/ProprietorShip/LLP/OPC)
 - Industry
 - YearsInOperation
 - AnnualTurnover
 - EmployeeCount
 - PrimaryBankID
 - KYCStatus
 - Status (Active/Inactive/Blacklisted)
- **Promoter**

- PromoterID
- BusinessID
- Name
- NationalIDRef
- ShareholdingPercent
- PersonalNetWorth
- CreditScore
- Status (Active/Inactive)
- **GroupCompany**
 - GroupID
 - ParentBusinessID
 - SubsidiaryBusinessID
 - Relationship (Subsidiary/Associate/JV)
 - ConsolidationRequired

4.3 Loan Application & Document Management Module

Features

- Accept structured loan applications with product type, purpose, and quantum details
- Manage document checklist, upload tracking, and verification status

Entities

- **LoanApplication**
 - ApplicationID
 - BusinessID
 - ProductType
(TermLoan/WorkingCapitalCC/OverdraftFacility/InvoiceFinancing/EquipmentLoan)
 - RequestedAmount
 - Tenure
 - Purpose

- ApplicationDate
- AssignedAnalystID
- Status
(Draft/Submitted/InReview/UnderwritingApproval/Sanctioned/Rejected/Disbursed)
- **ApplicationDocument**
 - DocID
 - ApplicationID
 - DocumentType
(AuditedFinancials/ITReturns/BankStatement/GSTReturns/MOA/COI/PropertyDocs)
 - FinancialYear
 - FilePath
 - UploadedDate
 - VerificationStatus (Pending/Verified/Deficient/Rejected)

4.4 Financial Analysis & Credit Underwriting Module

Features

- Capture multi-year financial data and compute key credit assessment ratios
- Prepare structured credit proposals with scorecard ratings and underwriter recommendations

Entities

- **FinancialStatement**
 - StatementID
 - ApplicationID
 - FinancialYear
 - Revenue
 - EBITDA
 - PAT
 - TotalAssets
 - TotalLiabilities

- NetWorth
- CurrentRatio
- DebtEquityRatio
- DSCR
- EnteredByID
- Status (Draft/Verified)
- **CreditProposal**
 - ProposalID
 - ApplicationID
 - AnalystID
 - ScorecardRating
 - RiskCategory (Low/Medium/High/Watchlist)
 - SuggestedAmount
 - SuggestedRate
 - Tenure
 - Conditions
 - AnalystRecommendation (Approve/Decline/Refer)
 - Status (Draft/Submitted/ApprovedByManager/Declined/Sanctioned)
- **UnderwritingDecision**
 - DecisionID
 - ProposalID
 - ManagerID
 - SanctionedAmount
 - ApprovedRate
 - Tenure
 - SpecialConditions
 - DecisionDate
 - Status (Approved/Declined/ConditionalApproval)

4.5 Collateral Management Module

Features

- Register collateral assets with type, ownership, and valuation details
- Track collateral coverage ratio against sanctioned facility and manage re-valuation cycles

Entities

- **CollateralRecord**
 - CollateralID
 - ApplicationID
 - AssetType (Property/Plant/Machinery/Receivables/Gold/Securities/FD)
 - Description
 - OwnerName
 - MarketValue
 - ForceValuePercent
 - RealisableValue
 - ValuationDate
 - ValuedByID
 - Status (Registered/Charged/Released/Disputed)
- **CollateralRevaluation**
 - RevaluationID
 - CollateralID
 - RevaluationDate
 - PreviousValue
 - NewValue
 - RevaluedByID
 - ChangePercent
 - Status (Completed/PendingApproval)

4.6 Facility Disbursement & Working Capital Management Module

Features

- Create sanctioned facility accounts and process drawdown requests against approved limits
- Track working capital utilisation, repayments, and available headroom

Entities

- **FacilityAccount**
 - FacilityID
 - ApplicationID
 - BusinessID
 - ProductType
 - SanctionedLimit
 - DisbursedAmount
 - OutstandingBalance
 - InterestRate
 - ExpiryDate
 - Status (Active/Expired/Closed/NPA)
- **Drawdown**
 - DrawdownID
 - FacilityID
 - Amount
 - Purpose
 - RequestDate
 - DisbursedDate
 - RepaymentDate
 - Status (Requested/Approved/Disbursed/Repaid/Overdue)
- **WorkingCapitalUtilisation**
 - UtilisationID
 - FacilityID

- Period
- DrawingPower
- CurrentUtilisation
- AvailableLimit
- UtilisationPercent
- RecordedDate

4.7 Covenant & Portfolio Monitoring Module

Features

- Define financial and non-financial covenants for each sanctioned facility
- Track covenant compliance, flag early warning signals, and manage watchlist classification

Entities

- **Covenant**
 - CovenantID
 - FacilityID
 - CovenantType (Financial/NonFinancial)
 - Description
 - ThresholdValue
 - MonitoringFrequency (Monthly/Quarterly/Annual)
 - Status (Active/Waived/Breached)
- **CovenantTracking**
 - TrackingID
 - CovenantID
 - Period
 - ActualValue
 - ThresholdValue
 - ComplianceStatus (Compliant/Breached/DataAwaited)
 - ReviewedByID

- ReviewDate
- **EarlyWarningSignal**
 - EWSID
 - FacilityID
 - SignalType
(UtilisationDrop/DelayedSubmission/CovenantBreach/SectorStress/Downgrade)
 - Severity (Green/Amber/Red)
 - DetectedDate
 - Status (Open/Actioned/Cleared)

4.8 Notifications & Alerts Module

Features

- Application status updates, document deficiency notices, and sanction confirmations for SME applicants
- Covenant breach alerts, EWS flags, facility renewal reminders, and drawdown overdue notifications for credit teams

Entities

- **Notification**
 - NotificationID
 - UserID
 - Message
 - Category (Application/Collateral/Facility/Covenant/EWS/Compliance)
 - Status (Unread/Read/Dismissed)
 - CreatedDate

5. Deployment Strategy

- **Local:** Angular/React frontend, Spring Boot/.NET Core backend, local DB
- **Production:** Cloud or on-premises with secure APIs and stateless services

6. Database Design

Tables:

- **User**(UserID, Name, Role, Email, Phone, BranchID, Status)
- **AuditLog**(AuditID, UserID, Action, EntityType, RecordID, Timestamp)
- **SMEBusiness**(BusinessID, BusinessName, RegistrationNumber, EntityType, Industry, YearsInOperation, AnnualTurnover, EmployeeCount, PrimaryBankID, KYCStatus, Status)
- **Promoter**(PromoterID, BusinessID, Name, NationalIDRef, ShareholdingPercent, PersonalNetWorth, CreditScore, Status)
- **GroupCompany**(GroupID, ParentBusinessID, SubsidiaryBusinessID, Relationship, ConsolidationRequired)
- **LoanApplication**(ApplicationID, BusinessID, ProductType, RequestedAmount, Tenure, Purpose, ApplicationDate, AssignedAnalystID, Status)
- **ApplicationDocument**(DocID, ApplicationID, DocumentType, FinancialYear, FilePath, UploadedDate, VerificationStatus)
- **FinancialStatement**(StatementID, ApplicationID, FinancialYear, Revenue, EBITDA, PAT, TotalAssets, TotalLiabilities, NetWorth, CurrentRatio, DebtEquityRatio, DSCR, EnteredByID, Status)
- **CreditProposal**(ProposalID, ApplicationID, AnalystID, ScorecardRating, RiskCategory, SuggestedAmount, SuggestedRate, Tenure, Conditions, AnalystRecommendation, Status)
- **UnderwritingDecision**(DecisionID, ProposalID, ManagerID, SanctionedAmount, ApprovedRate, Tenure, SpecialConditions, DecisionDate, Status)
- **CollateralRecord**(CollateralID, ApplicationID, AssetType, Description, OwnerName, MarketValue, ForceValuePercent, RealisableValue, ValuationDate, ValuedByID, Status)
- **CollateralRevaluation**(RevaluationID, CollateralID, RevaluationDate, PreviousValue, NewValue, RevaluedByID, ChangePercent, Status)
- **FacilityAccount**(FacilityID, ApplicationID, BusinessID, ProductType, SanctionedLimit, DisbursedAmount, OutstandingBalance, InterestRate, ExpiryDate, Status)
- **Drawdown**(DrawdownID, FacilityID, Amount, Purpose, RequestDate, DisbursedDate, RepaymentDate, Status)
- **WorkingCapitalUtilisation**(UtilisationID, FacilityID, Period, DrawingPower, CurrentUtilisation, AvailableLimit, UtilisationPercent, RecordedDate)
- **Covenant**(CovenantID, FacilityID, CovenantType, Description, ThresholdValue, MonitoringFrequency, Status)

- **CovenantTracking**(TrackingID, CovenantID, Period, ActualValue, ThresholdValue, ComplianceStatus, ReviewedByID, ReviewDate)
- **EarlyWarningSignal**(EWSID, FacilityID, SignalType, Severity, DetectedDate, Status)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **SME Applicant Portal:** Application form wizard, document upload checklist, application tracker, facility statement
- **Credit Analyst Workbench:** Financial data entry, ratio calculator, credit proposal builder, document review queue
- **Underwriting Manager Console:** Proposal review queue, sanction decision form, maker-checker approval panel
- **Relationship Manager Dashboard:** Client facility portfolio, covenant compliance tracker, renewal pipeline, EWS flag board
- **Collateral Management Console:** Asset registration form, valuation tracker, coverage ratio calculator, re-valuation scheduler
- **Admin Console:** Loan product configurator, scorecard builder, covenant template library, user role management

8. Non-Functional Requirements

- **Performance:** Support 10,000+ concurrent users across SME applicants, credit analysts, and relationship managers
- **Security:** Financial data access scoped by role, collateral record encryption, RBAC, and full audit trails for credit decisions
- **Scalability:** Multi-branch and multi-product architecture for commercial banking and NBFC deployments
- **Availability:** 99.95% uptime with credit proposal and facility management modules prioritised for high availability
- **Maintainability:** Configurable underwriting scorecards, covenant templates, and product parameters without code changes
- **Observability:** Real-time SME portfolio dashboards, covenant compliance tracking, and EWS signal monitoring

9. Assumptions & Constraints

- Phase 1 excludes integration with external credit bureau APIs, GST analytics platforms, and valuation agency portals
- Financial ratios are computed from manually entered financial statement data; automated extraction is deferred to Phase 2
- All notifications are in-app only; email and SMS integrations for SME applicants are out of Phase 1 scope
- Collateral valuation is entered by internal evaluators; integration with third-party valuation agencies is deferred
- Fully implementable using standard Java Spring Boot / ASP.NET Core, Angular/React, and RDBMS tooling